SAFETY DATA SHEET		Revision Date: 25.10.2024
		Print Date:26.10.2024
		SDS Number: 000000273004
mirage 146a™ biofunctional ™ Trademark, Ashland or its subsidiaries, registered in various countries 882504		Version: 4.0

According to REACH Regulation (EC) No 1907/2006, as amended by UK REACH Regulations SI 2019/758 - SDSGHS_GB

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name : mirage 146a™ biofunctional
Description : biofunctional
™ Trademark, Ashland or its subsidiaries, registered in various countries

1.2 Relevant identified uses of the substance or mixture and uses advised against

Recommended use : biofunctional

1.3 Details of the supplier of the safety data sheet	**1.4 Emergency telephone**
Ashland Specialties UK Ltd 30 Old Bailey London EC4M 7AU United Kingdom 001-614-790-3333 (customer service USA) Company: ISP Microcaps (UK) Ltd 8 – 9 Stepnell Reach, 541 Blandford Rd Poole, Dorset, BH16 5BW, England 01202 627352 Techwax Ltd 4B Whinbank Park, Whinbank Road Newton Aycliffe Industrial Park, Newton Aycliffe Co Durham, DL5 6AY, United Kingdom 01325 303084 EHSProductSafety@ashland.com	1-800-ASHLAND (1-800-274-5263), or contact your local emergency telephone number at 111 - ENG, SCO, CYM. **Regulatory information** 1-614-790-3333 (customer service) **Product Information** 001-614-790-3333 (customer service USA)

SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008) as amended by GB-CLP Regulation, UK SI 2019/720, and UK SI 2020/1567)

Not a hazardous substance or mixture.

2.2 Label elements

Labeling (REGULATION (EC) No 1272/2008) as amended by GB-CLP Regulation, UK SI 2019/720, and UK SI 2020/1567)

No hazard pictogram, no signal word, no hazard statement(s), no precautionary statement(s) required.

Additional Labeling

EUH210 Safety data sheet available on request.

2.3 Other hazards

This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

SECTION 3: Composition/information on ingredients

3.2 Mixtures

Chemical nature : organic

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration (% w/w)
2-phenoxyethanol	122-99-6 204-589-7 603-098-00-9 01-2119488943-21-0020	Acute Tox. 4; H302 Eye Dam. 1; H318 STOT SE 3; H335 (Respiratory system)	>= 1 - < 2,5
Substances with a workplace exposure limit :			

glycerol	56-81-5 200-289-5 01-2119471987-18-xxxx		>= 25 - < 40

For explanation of abbreviations see section 16.

SECTION 4: First aid measures

4.1 Description of first-aid measures

General advice : Do not leave the victim unattended.

If inhaled : If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a physician.

In case of skin contact : Wash off with soap and plenty of water.

In case of eye contact : Remove contact lenses.
Protect unharmed eye.
If eye irritation persists, consult a specialist.

If swallowed : Keep respiratory tract clear.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.

4.2 Most important symptoms and effects, both acute and delayed

None known.

4.3 Indication of any immediate medical attention and special treatment needed

Treatment : Treat symptomatically.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Use water spray, alcohol-resistant foam, dry chemical or carbon dioxide.

5.2 Special hazards arising from the substance or mixture

Specific hazards during fire fighting	:	No information available.
Hazardous combustion products	:	Carbon dioxide (CO2) Carbon monoxide Nitrogen oxides (NOx) Aldehydes

5.3 Advice for firefighters

Special protective equipment for fire-fighters	:	In the event of fire, wear self-contained breathing apparatus.
Specific extinguishing methods	:	Standard procedure for chemical fires.
Further information	:	Standard procedure for chemical fires. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions	:	Use personal protective equipment. Avoid dust formation.

6.2 Environmental precautions

Environmental precautions	:	Do not allow uncontrolled discharge of product into the environment.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up	:	Pick up and arrange disposal without creating dust. Sweep up and shovel. Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

See sections: 7, 8, 11, 12 and 13.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advice on safe handling : Avoid dust formation.
Avoid inhalation, ingestion and contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.

Advice on protection against fire and explosion : Provide appropriate exhaust ventilation at places where dust is formed.

Hygiene measures : General industrial hygiene practice.

7.2 Conditions for safe storage, including any incompatibilities

Requirements for storage areas and containers : Electrical installations / working materials must comply with the technological safety standards.

Further information on storage conditions : Keep container tightly closed in a dry and well-ventilated place. Store at room temperature in the original container. Keep in a dry, cool place. Keep away from direct sunlight.

Advice on common storage : No materials to be especially mentioned.

Further information on storage stability : No decomposition if stored and applied as directed.

7.3 Specific end use(s)

Specific use(s) : No data available

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Occupational Exposure Limits

Components	CAS-No.	Value type (Form of exposure)	Control parameters	Basis
glycerol	56-81-5	TWA (Mist)	10 mg/m3	GB EH40
	Further information: Where no specific short-term exposure limit is listed, a figure three times the long-term exposure limit should be used.			

Derived No Effect Level (DNEL):

Substance name	End Use	Routes of exposure	Potential health effects	Value
2-phenoxyethanol	GEN POP - General population	Dermal	Long-term systemic effects	20,83 mg/kg
	Remarks:RD TOX - Repeated dose toxicity			
	WRKS - Workers	Dermal	Long-term systemic effects	20,83 meq/kg
	WRKS - Workers	Inhalation	Long-term systemic effects	5,7 mg/m3
	WRKS - Workers	Inhalation	Long-term local effects	5,7 mg/m3
	Consumers	Dermal	Long-term systemic effects	10,42 mg/kg
	Consumers	Inhalation	Long-term systemic effects	2,41 mg/m3
	Consumers	Oral	Long-term systemic effects	9,23 mg/m3
	Consumers	Oral	Acute systemic effects	9,23 mg/m3

Predicted No Effect Concentration (PNEC):

Substance name	Environmental Compartment	Value
2-phenoxyethanol	Aquatic (freshwater)	0,943 mg/l
	Sewage treatment plant	24,8 mg/l
	Aquatic (intermit. releases)	3,44 mg/l
	Aquatic (marine water)	0,0943 mg/l
	Sediment (freshwater)	7,2366 mg/kg
	Sediment (marine water)	0,7237 mg/kg
	Soil	1,26 mg/kg

8.2 Exposure controls

Engineering measures

Provide sufficient mechanical (general and/or local exhaust) ventilation to maintain exposure below exposure guidelines (if applicable) or below levels that cause known, suspected or apparent adverse effects.

Personal protective equipment

Eye/face protection : Safety glasses

Hand protection
Material : butyl-rubber
Break through time : 480 min

Glove thickness	:	> 0,5 mm
Remarks	:	The exact break through time can be obtained from the protective glove producer and this has to be observed. Gloves should be discarded and replaced if there is any indication of degradation or chemical breakthrough.
Skin and body protection	:	Work uniform or laboratory coat.
Respiratory protection	:	In the case of dust or aerosol formation use respirator with an approved filter within the capabilities of the respirator/filter combination. Where concentrations are above recommended limits or are unknown, or a cartridge type respirator is not adequate, wear a positive-pressure supplied-air respirator.

Environmental exposure controls

Soil	:	The product should not be allowed to enter drains, water courses or the soil.
Water	:	If the product contaminates rivers and lakes or drains inform respective authorities.

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance	:	Crystalline product
Color	:	amber
Odor	:	characteristic
Odor Threshold	:	No data available
pH	:	5,5 - 6,5
Melting point/freezing point	:	not determined
Boiling point/boiling range	:	not determined

Flash point	:	175 °C
Evaporation rate	:	Not applicable
Flammability (solid, gas)	:	not determined
Upper explosion limit / Upper flammability limit	:	not determined
Lower explosion limit / Lower flammability limit	:	not determined
Vapor pressure	:	Not applicable
Relative vapor density	:	Not applicable
Relative density	:	No data available
Density	:	not determined
Solubility(ies)		
Water solubility	:	not determined
Solubility in other solvents	:	No data available
Partition coefficient: n-octanol/water	:	not determined
Decomposition temperature	:	No data available
Viscosity		
Viscosity, dynamic	:	Not applicable
Viscosity, kinematic	:	Not applicable
Oxidizing properties	:	Not applicable

9.2 Other information

Flammability (liquids)	:	not determined
Molecular weight	:	No data available

Self-ignition : not determined

SECTION 10: Stability and reactivity

10.1 Reactivity

No dangerous reaction known under conditions of normal use.

10.2 Chemical stability

Stable under normal conditions.

10.3 Possibility of hazardous reactions

Hazardous reactions : Dust may form explosive mixture in air.

10.4 Conditions to avoid

Conditions to avoid : Protect from frost, heat and sunlight.

10.5 Incompatible materials

Materials to avoid : Oxidizing agents

10.6 Hazardous decomposition products

Hazardous decomposition products : Carbon monoxide
Carbon dioxide (CO2)

SECTION 11: Toxicological information

11.1 Information on toxicological effects

Information on likely routes of exposure :
Inhalation
Eye contact
Skin contact
Ingestion

Acute toxicity

Not classified due to lack of data.

Product:

Acute oral toxicity : Acute toxicity estimate: > 2.000 mg/kg
Method: Calculation method

Components:

2-PHENOXYETHANOL:

Acute oral toxicity	:	LD50 (Rat, male): 1.394 mg/kg
Acute inhalation toxicity	:	Assessment: No adverse effect has been observed in acute inhalation toxicity tests.
Acute dermal toxicity	:	LD50 (Rat): 14.391 mg/kg

GLYCERINE:

Acute oral toxicity	:	LD50 (Rat): 12.600 mg/kg Method: OECD Test Guideline 401
Acute inhalation toxicity	:	LC50 (Rat): > 570 mg/m3 Exposure time: 1 h Test atmosphere: vapor Assessment: Not classified as acutely toxic by inhalation under GHS.
Acute dermal toxicity	:	LD50 (Rabbit): > 10 g/kg

Skin corrosion/irritation

Based on available data, the classification criteria are not met.

Product:

Species	:	reconstructed human epidermis (RhE)
Result	:	No skin irritation
Species	:	Human
Exposure time	:	48 h
Method	:	human patch test
Result	:	No skin irritation
Species	:	Human
Exposure time	:	24 h
Method	:	Human repeat insult patch test (HRIPT)
Result	:	No skin irritation

Components:

2-PHENOXYETHANOL:

Species	:	Rabbit
Result	:	No skin irritation

GLYCERINE:

Result	:	No skin irritation

Serious eye damage/eye irritation

Based on available data, the classification criteria are not met.

Product:

Result	:	No eye irritation
Method	:	EpiOcular
Result	:	No eye irritation
Method	:	Chorioallantoic membrane vascularization assay
Result	:	No eye irritation
Method	:	Reconstructed Human Corneal Model
Result	:	No eye irritation

Components:

2-PHENOXYETHANOL:

Species	:	Rabbit
Result	:	Corrosive

GLYCERINE:

Result	:	Slight, transient irritation

Respiratory or skin sensitization

Skin sensitization

Based on available data, the classification criteria are not met.

Respiratory sensitization

Not classified due to lack of data.

Product:

Test Type	:	Human Repeat Insult Patch Testing (HRIPT)
Routes of exposure	:	Dermal
Species	:	Humans
Method	:	Human repeat insult patch test (HRIPT)

Result : Not a skin sensitizer.

Components:

2-PHENOXYETHANOL:

Species	:	Guinea pig
Assessment	:	Does not cause skin sensitization.
Method	:	OECD Test Guideline 406

Germ cell mutagenicity

Not classified due to lack of data.

Product:

Genotoxicity in vitro : Test Type: Ames test
Test system: Salmonella typhimurium
Metabolic activation: with and without metabolic activation
Result: negative

Test Type: Ames test
Test system: Escherichia coli
Metabolic activation: with and without metabolic activation
Result: negative

Components:

2-PHENOXYETHANOL:

Genotoxicity in vitro	:	Test Type: Ames test Test system: Salmonella typhimurium Metabolic activation: with and without metabolic activation Result: negative

Carcinogenicity

Not classified due to lack of data.

Reproductive toxicity

Not classified due to lack of data.

Components:

2-PHENOXYETHANOL:

Effects on fetal development	:	Test Type: Pre-natal Species: Rat Application Route: Oral General Toxicity Maternal: NOAEL: ca. 300 mg/kg bw/day

		Method: OPPTS 870.3700

STOT-single exposure

Not classified due to lack of data.

Components:

2-PHENOXYETHANOL:

Assessment	:	May cause respiratory irritation.

STOT-repeated exposure

Not classified due to lack of data.

Repeated dose toxicity

Components:

2-PHENOXYETHANOL:

Species	:	Rat, male and female
NOAEL	:	369 mg/kg
Application Route	:	Oral
Method	:	OECD Test Guideline 408

Species	:	Rabbit, male and female
NOAEL	:	500 mg/kg
Application Route	:	Dermal

Aspiration toxicity

Not classified due to lack of data.

Further information

Product:

Remarks : No data available

SECTION 12: Ecological information

12.1 Toxicity

Components:

2-PHENOXYETHANOL:

Toxicity to fish	:	LC50 (Pimephales promelas (fathead minnow)): 337 - 352 mg/l Exposure time: 96 h

		Test Type: flow-through test
Toxicity to daphnia and other aquatic invertebrates	:	EC50 (Daphnia magna (Water flea)): > 500 mg/l Exposure time: 48 h Test Type: static test Method: OECD Test Guideline 202
Toxicity to algae/aquatic plants	:	NOEC (Desmodesmus subspicatus (green algae)): > 500 mg/l End point: Growth inhibition Exposure time: 72 h Test Type: static test
Toxicity to fish (Chronic toxicity)	:	NOEC: 23 mg/l Exposure time: 34 d Species: Pimephales promelas (fathead minnow) Test Type: flow-through test Method: OECD Test Guideline 210
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	:	NOEC: 9,43 mg/l End point: Reproduction Test Exposure time: 21 d Species: Daphnia Test Type: semi-static test Method: OECD Test Guideline 211

GLYCERINE:

Toxicity to fish	:	LC50 (Oncorhynchus mykiss (rainbow trout)): 54.000 mg/l Exposure time: 96 h Test Type: static test
Toxicity to daphnia and other aquatic invertebrates	:	LC50 (Daphnia magna (Water flea)): 1.995 mg/l Exposure time: 48 h

12.2 Persistence and degradability

Components:

2-PHENOXYETHANOL:

Biodegradability	:	Result: Readily biodegradable. Biodegradation: 99 % Exposure time: 28 d Method: OECD Test Guideline 301F

GLYCERINE:

Biodegradability	:	Result: Readily biodegradable. Biodegradation: 94 %

12.3 Bioaccumulative potential

Components:

2-PHENOXYETHANOL:

Partition coefficient: n-octanol/water	:	log Pow: 1,16

GLYCERINE:

Partition coefficient: n-octanol/water	:	log Pow: -1,76

12.4 Mobility in soil

No data available

12.5 Results of PBT and vPvB assessment

Product:

Assessment	:	This substance/mixture contains no components considered to be either persistent, bioaccumulative and toxic (PBT), or very persistent and very bioaccumulative (vPvB) at levels of 0.1% or higher.

Components:

2-PHENOXYETHANOL:

Assessment	:	Substance is not persistent, bioaccumulative, and toxic (PBT).
	:	Substance is not very persistent and very bioaccumulative (vPvB).

12.6 Other adverse effects

Product:

Endocrine disrupting potential	:	This substance/mixture does not contain components considered to have endocrine disrupting properties for environment according to UK REACH Article 57(f).
Additional ecological information	:	No data available

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Contaminated packaging : Empty containers should be taken to an approved waste handling site for recycling or disposal.

SECTION 14: Transport information

14.1 UN number

ADR : Not regulated as a dangerous good

RID : Not regulated as a dangerous good

IMDG : Not regulated as a dangerous good

IATA : Not regulated as a dangerous good

14.2 UN proper shipping name

ADR : Not regulated as a dangerous good

RID : Not regulated as a dangerous good

IMDG : Not regulated as a dangerous good

IATA : Not regulated as a dangerous good

14.3 Transport hazard class(es)

ADR : Not regulated as a dangerous good

RID : Not regulated as a dangerous good

IMDG : Not regulated as a dangerous good

IATA : Not regulated as a dangerous good

14.4 Packing group

ADR : Not regulated as a dangerous good

RID : Not regulated as a dangerous good

IMDG : Not regulated as a dangerous good

IATA (Cargo) : Not regulated as a dangerous good

IATA (Passenger) : Not regulated as a dangerous good

14.5 Environmental hazards

Not regulated as a dangerous good

14.6 Special precautions for user

Not applicable

14.7 Transport in bulk according to Annex II of Marpol and the IBC Code

Not applicable for product as supplied.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Relevant EU provisions transposed through retained EU law

UK REACH List of restrictions (Annex 17)	:	Not applicable
UK REACH Candidate list of substances of very high concern (SVHC) for Authorisation	:	Not applicable
The Persistent Organic Pollutants Regulations (retained Regulation (EU) 2019/1021 as amended for Great Britain)	:	Not applicable
Regulation (EC) on substances that deplete the ozone layer	:	Not applicable
UK REACH List of substances subject to authorisation (Annex XIV)	:	Not applicable
GB Export and import of hazardous chemicals - Prior Informed Consent (PIC) Regulation	:	Not applicable
Control of Major Accident Hazards Regulations 2015 (COMAH)		Not applicable

The ingredients of this product are reported in the following inventories:

TCSI	:	On the inventory, or in compliance with the inventory
AIIC	:	Not in compliance with the inventory
DSL	:	This product contains one or several components that are not on the Canadian DSL and have annual quantity limits.

ISHL	:	Not in compliance with the inventory
ENCS	:	Personal care
KECI	:	Not in compliance with the inventory
PICCS	:	Not in compliance with the inventory
IECSC	:	On the inventory, or in compliance with the inventory
TSCA	:	This product is regulated under the United States Food and Drug Act (FDA).

15.2 Chemical Safety Assessment

No data available

SECTION 16: Other information

Full text of H-Statements

H302	:	Harmful if swallowed.
H318	:	Causes serious eye damage.
H335	:	May cause respiratory irritation.

Full text of other abbreviations

Acute Tox.	:	Acute toxicity
Eye Dam.	:	Serious eye damage
STOT SE	:	Specific target organ toxicity - single exposure
GB EH40	:	UK. EH40 WEL - Workplace Exposure Limits
GB EH40 / TWA	:	Long-term exposure limit (8-hour TWA reference period)

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health

Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - substance of very high concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative

Further information

Other information	:	This sds has been prepared by Ashland. (http://www.ashland.com)
Sources of key data used to compile the Material Safety Data Sheet	:	Ashland internal data including own and sponsored test reports European Union Law with content from the Official Journal of the European Union. European Chemicals Agency; the EU authority implementing the EU's chemicals legislation for companies. The German Water Hazard Classes. ReachCentrum; a series of support services to help comply with REACH regulations. The European Commission; proposing legislation, administering and implementing EU policies, and enforcing EU law. The UNECE administers regional agreements implementing harmonised classification for labelling (GHS) and transport. Cefic, the European Chemical Industry Council. ESIS European Chemical Substances Information System

Items where changes have been made to the previous version are highlighted in the body of this document by two vertical lines.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

GB / EN